Unit: Unit 1: Ratio, Rate, Percentage
Topic: Rates Variation

Name: ____________
Date: ________

Foundation (Jalapeno)

Q1. A spaceship travels 240 km in 4 hours. What is its unit rate?
(A) 50 km/h
(B) 60 km/h
(C) 70 km/h
(D) 80 km/h
(E) 90 km/h

Q2. A scientist mixes 2 liters of a chemical with 5 liters of water. What is the ratio of chemical to total mixture?
(A) 1 : 3
(B) 1 : 4
(C) 1 : 5
(D) 1 : 6
(E) 1 : 7

Q3. The cost of launching a satellite is directly proportional to its mass. If a 500 kg satellite costs

$$120,000$$

to launch, how much will it cost to launch a 750 kg satellite?
(A)
$$180,000$$
(B)
$$216,000$$
(C)
$$240,000$$
(D)
$$270,000$$
(E)
$$300,000$$

Q4. A rocket travels at a rate of 200 m/s. How far will it travel in 10 seconds?
(A) 1500 m
(B) 1800 m
(C) 2000 m
(D) 2200 m
(E) 2500 m

Q5. The time it takes for a spacecraft to orbit the Earth is inversely proportional to its velocity. If a spacecraft orbits the Earth in 90 minutes at a velocity of 7.8 km/s, how long will it take to orbit the Earth at a velocity of 10.2 km/s?
(A) 60 minutes
(B) 70 minutes
(C) 75 minutes
(D) 80 minutes
(E) 90 minutes

Q6. A lab experiment requires a ratio of 3 : 5 of two chemicals. If 21 liters of the first chemical are used, how many liters of the second chemical are needed?
(A) 30 liters
(B) 35 liters
(C) 40 liters
(D) 45 liters
(E) 50 liters

Q7. A telescope can view a planet at a distance of 50 million km in 10 seconds. How far away can it view the planet in 5 seconds?
(A) 20 million km
(B) 25 million km
(C) 30 million km
(D) 40 million km
(E) 50 million km

Q8. A spacecraft travels 300 km in 5 hours. What is its average speed in km/h?
(A) 50 km/h
(B) 60 km/h
(C) 70 km/h
(D) 80 km/h
(E) 90 km/h

Open Ended Questions (FRQ)

Q9. A lab experiment requires a mixture of two chemicals in a ratio of 2 : 3. If 16 liters of the first chemical are used, how many liters of the second chemical are needed? Show your work and calculate the total volume of the mixture.

Q10. The distance a spacecraft travels is directly proportional to the square of its velocity. If a spacecraft travels 1500 km at a velocity of 20 m/s, how far will it travel at a velocity of 30 m/s? Show your work and explain your reasoning.

Chef's Tips

- Emphasize the importance of units and conversions in rate problems
- Encourage students to use visual models to represent direct and inverse variation
- Remind students to check their work and calculate carefully